# Sampling

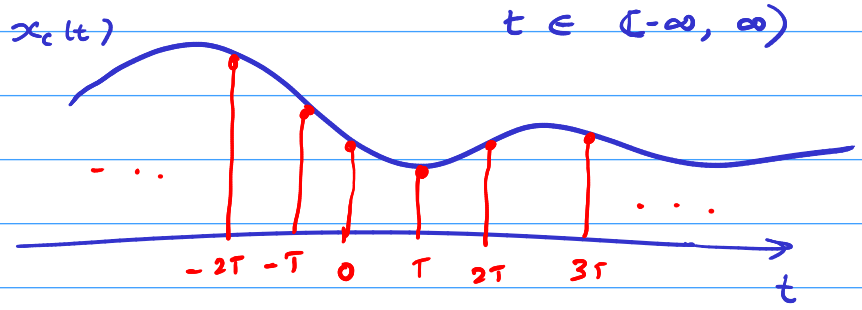
Continuous time signal $x_c(t)$

time

$t \rightarrow$ continuous variable

$t \in \mathbb{R}$

$t \in (-\infty, \infty)$



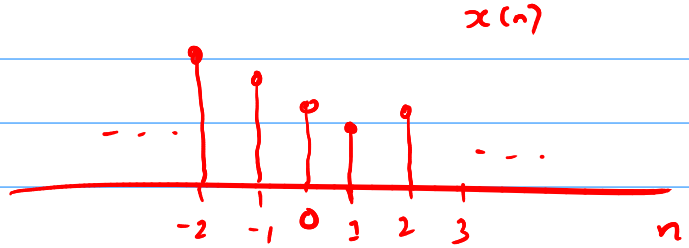

Sampling Theory (convert CT signal to DT signal and vice versa)

Design equivalent DT system for corresponding CT system

## Periodic Sampling

Take samples of $x_c(t)$ at interval of T seconds

$$x(n) = x_c(t)\Big|_{@\, t = nT}, \quad n \in \mathbb{Z}$$




Sampling Interval : T seconds

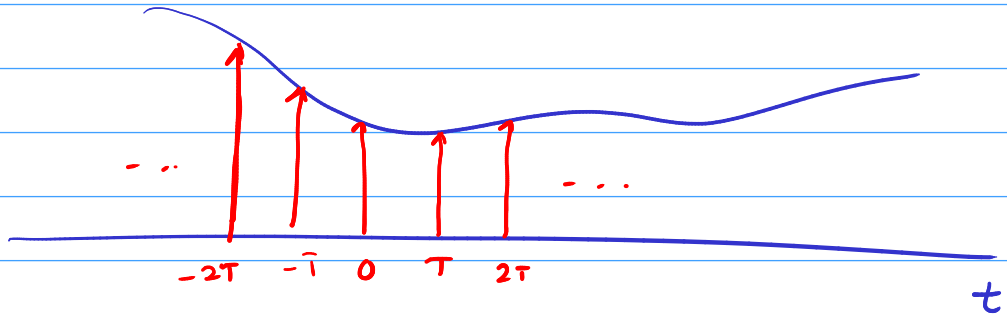
Sampling frequency $f_s = \frac{1}{T}$ Hz

Sampling angular freq $\Omega_s = \frac{2\pi}{T}$ radian/sec

} Sampling Parameters

Define an intermediate signal (CT)

$x_s(t)$




$x_s(t)$ = Sequence of dirac delta impulses

located @ nT

Area under impulse @ location nT is

CT signal $x_c(nT)$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\,\delta(t-nT)$$

$\Downarrow$

Sampled CT signal

Can we reconstruct $x_c(t)$ using $x_s(t)$ (or $x(n)$)?

We need to relate Fourier transform of

$x_c(t)$ and $x_s(t)$

CTFT (Continuous time Fourier Transform)

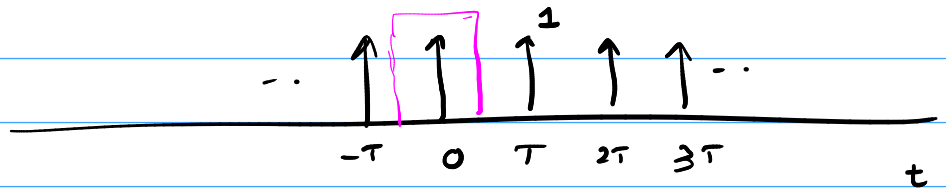
$$X_c(\Omega) = \int_{-\infty}^{\infty} x_c(t)\, e^{-j\Omega t}\, dt$$

$\Omega$ => angular frequency variable for CT signal

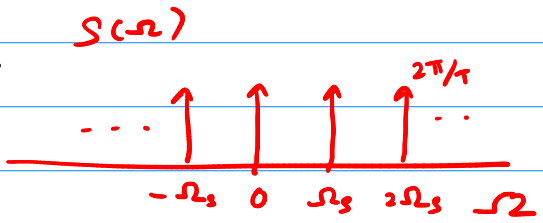
(radians/second)

Need to find $X_s(\Omega)$ for Sampled signal $x_s(t)$ — Spectrum — Intermediate

Consider Periodic Impulse train $s(t)$




What is Spectrum of $S(\Omega)$ ?




Note $x_s(t) = x_c(t) \cdot s(t)$

↑ Multiplication

Now $X_s(\Omega) = \frac{1}{2\pi}\left[X_c(\Omega) * S(\Omega)\right]$

↓ Convolution

$x(t) * \delta(t) = x(t)$

$x(t) * \delta(t-t_0) = x(t-t_0)$

$= \frac{1}{2\pi} X_c(\Omega) * \sum_{k=-\infty}^{\infty} \frac{2\pi}{T}\delta(\Omega - k\Omega_s)$

$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(\Omega) * \delta(\Omega - k\Omega_s)$

$$X_s(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$$

Spectrum of Sampled Signal $x_s(t)$

Sum of shifted copies of spectrum of Original Signal $x_c(t)$

# Spectrum of Impulse train

Impulse train $s(t)$ is periodic with period $T$

Fundamental freq. $\frac{2\pi}{T} = \Omega_s$

$k^{th}$ harmonic $k\Omega_s = k\frac{2\pi}{T}$

CT Fourier Series

$$s(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\Omega_s t}$$

($k^{th}$ harmonic)

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} s(t)\, e^{-jk\Omega_s t}\, dt$$

$$= \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)\, e^{-jk\Omega_s t}\, dt$$

$\Downarrow f(t)$

$$a_k = \frac{1}{T} f(0) \qquad f(0) = 1$$

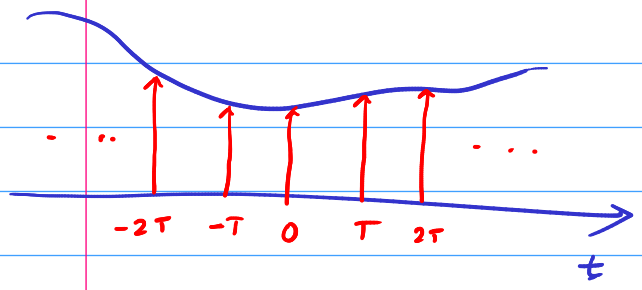
$$\boxed{a_k = 1/T}$$

$$s(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\Omega_s t}$$

$$e^{jk\Omega_s t} \xrightarrow{\text{CTFT}} 2\pi\,\delta(\Omega - k\Omega_s)$$

$$S(\Omega) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{T}\,\delta(\Omega - k\Omega_s)$$

$x_c(t)$




$$x_s(t) = \sum_n x(nT)\,\delta(t-nT)$$

Sampling interval $T$ sec

angular Sampling freq $\Omega_s = \frac{2\pi}{T}$ rad/sec

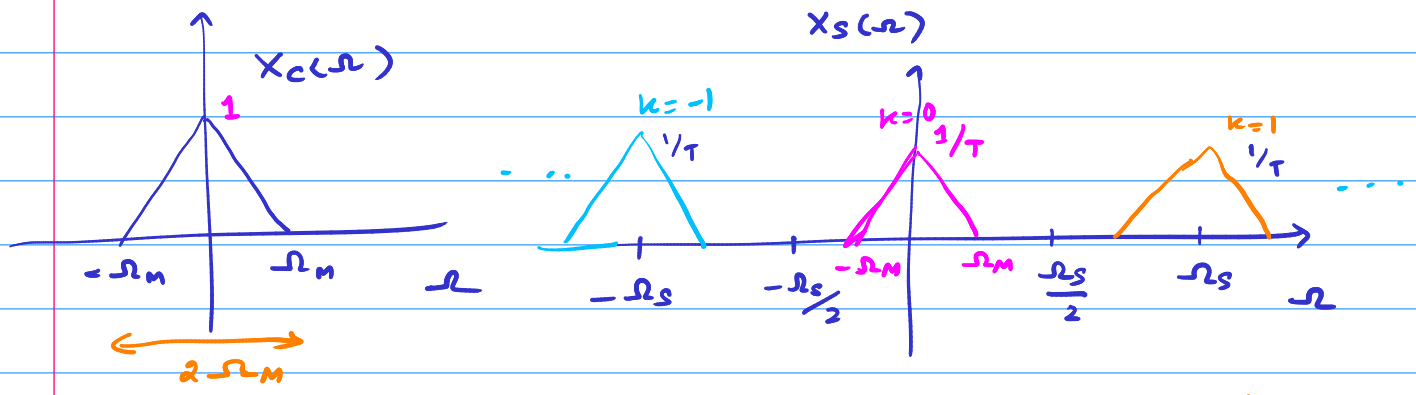
$X_c(\Omega) \Rightarrow$ spectrum of $x_c(t)$

$$X_s(\Omega) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X_c(\Omega - k\Omega_s)$$

Suppose $X_c(\Omega)$ is bandlimited (non zero in a finite interval)

$X_c(\Omega) = 0$ if $|\Omega| > \Omega_M$

$\Omega_M \Rightarrow$ Maximum freq. present in signal $x_c(t)$




Case A

$\Omega_s > 2\Omega_M$

shifted copies of original spectrum do not overlap

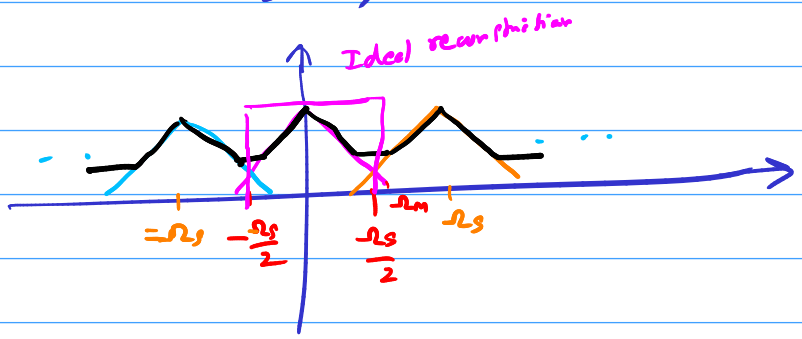
In this case $X_c(\Omega) = T\,X_s(\Omega)$ for all $|\Omega| \le \Omega_s/2$

Then there is no distortion due to sampling

We can recover original signal from sampled signal.

Case B $\quad \Omega_s < 2\Omega_M$

$X_s(\Omega)$




This is aliasing distortion

Perfect reconstruction NOT possible

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Connection between discrete time and continuous time spectra

DT signal

$x(n) = x_c(t)\big|_{@\, t=nT} = x_c(nT)$

DT samples of $x_c(t)$

$x_c(t) \Rightarrow$ CT signal

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t-nT)$$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(n)\,\delta(t-nT)$$

DT signal $\{x(n)\} \rightarrow$ Spectrum

DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

Spectrum of $x_s(t)$

$$X_s(\Omega) = \int_{-\infty}^{\infty} x_s(t)\, e^{-j\Omega t}\, dt$$

$$= \int_{-\infty}^{\infty} \left(\sum_n x(n)\,\delta(t-nT)\right) e^{-j\Omega t}\, dt$$

$$= \sum_{n=-\infty}^{\infty} x(n) \underbrace{\int_{-\infty}^{\infty} \delta(t-nT)\, e^{-j\Omega t}\, dt}_{e^{-j\Omega nT}}$$

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\Omega T n}$$ $\Omega \Rightarrow$ CT freq.

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$ $\omega \Rightarrow$ DT freq.

$T = \frac{2\pi}{\Omega_s}$

$\omega = \Omega T \Rightarrow$ Connection between DT freq. $\omega$ & CT freq. $\Omega$ (with sampling interval $T$)

$\omega = \Omega \frac{2\pi}{\Omega_s}$

$X(\omega) = X_s(\Omega T)$

CT freq. $\Omega$ | DT freq. $\omega = \Omega T = \Omega \cdot \frac{2\pi}{\Omega_s}$

$\Omega = 0$ | $\omega = 0$

$\Omega = \frac{\Omega_s}{4}$ | $\omega = \pi/2$

$\Omega = \Omega_s/2$ | $\omega = \pi$

$\Omega = \Omega_s$ | $\omega = 2\pi$

$\Omega = \Omega_s + \frac{\Omega_s}{4}$ | $\omega = 2\pi + \pi/2 = \pi/2$

$e^{j\Omega_0 t}$ —Periodic Sampling @ $nT$→ $e^{j\Omega_0 nT} = e^{j\Omega_0 \frac{2\pi}{\Omega_s} n} = e^{j\omega_0 n}$

$\omega_0 = \Omega_0 \frac{2\pi}{\Omega_s}$

## Properties of Sampling

(impulse train)

1. Spectra of sampled signal $x_s(t)$ is always periodic with period $\Omega_s$

$$X_s(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$$

$$X_s(\Omega+\Omega_s) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega+\Omega_s - k\Omega_s)$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega - (k-1)\Omega_s)$$

$l = k-1$

$$= \frac{1}{T}\sum_{l=-\infty}^{\infty} X_c(\Omega - l\Omega_s)$$

$$\boxed{X_s(\Omega+\Omega_s) = X_s(\Omega)} \quad \text{for any } \Omega.$$

2. Shannon's Sampling Theorem.

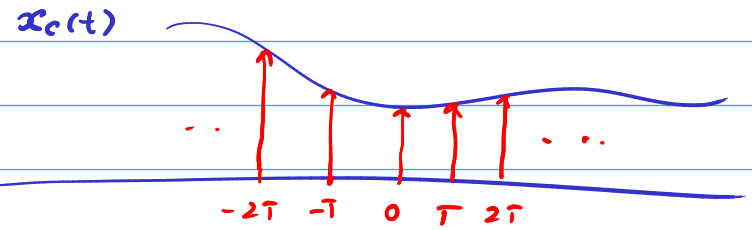
$$\boxed{\text{Nyquist rate } \Omega_N = 2\Omega_M}$$

$\Omega_M \Rightarrow$ max freq. present in $x_c(t)$

If sampling freq. is $\Omega_s > 2\Omega_M$ (= $\Omega_N$)

then original signal $x_c(t)$ can be perfectly recovered from sampled signal $x_s(t)$

# Sampling, Reconstruction & Aliasing




$\Omega_s = \frac{2\pi}{T}$

Spectrum

$X_c(\Omega)$

$X_s(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$

$x(n) = x_c(nT) \; ; \; n \in \mathbb{Z}$

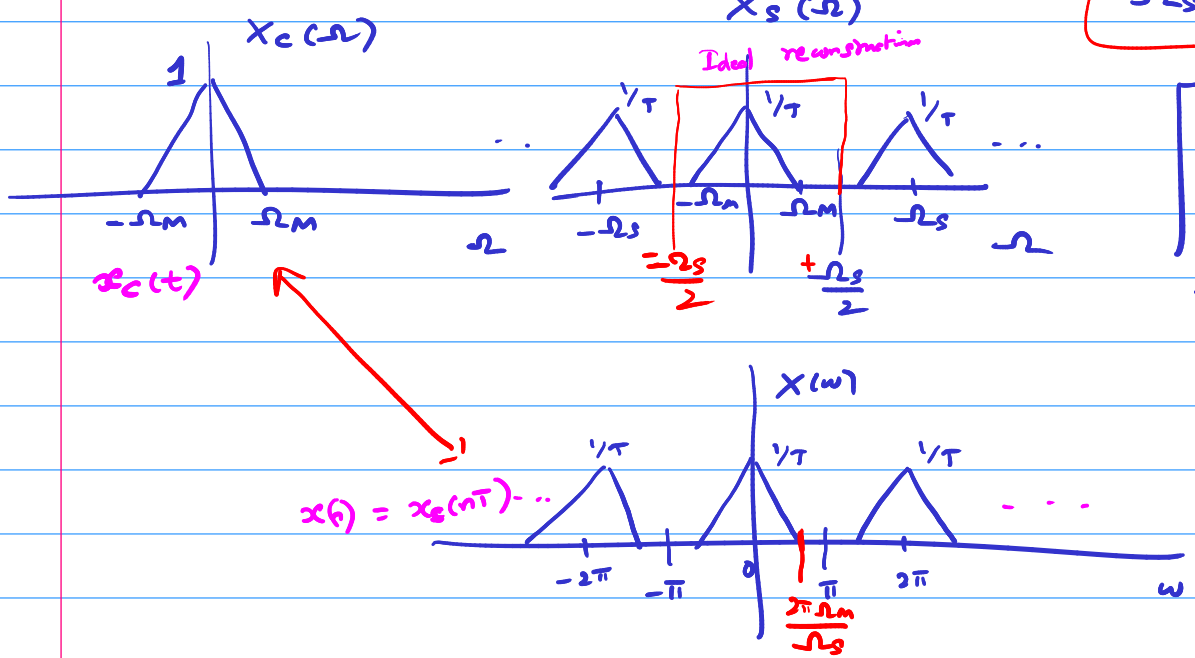
$x_s(t) = \sum_{n=-\infty}^{\infty} x(n)\,\delta(t-nT)$

$X(\omega) = X_s(\omega = \Omega T)$

$= X_s\left(\omega = \frac{\Omega 2\pi}{\Omega_s}\right)$

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Illustration



# Reconstructing CT signal from DT samples

$x_c(t)$ is bandlimited

$\Omega_m$ is max freq. in spectrum of $X_c(\Omega)$

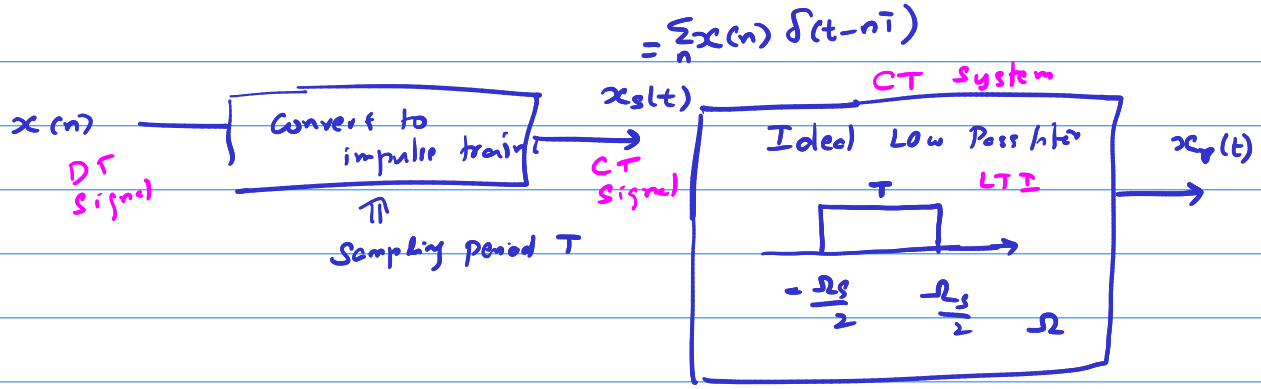
$X_c(\Omega) = 0$ if $|\Omega| > \Omega_m$

$x(n) = x_c(nT)$ ; $n \in \mathbb{Z}$

$T$ sampling interval

$\Omega_s = \frac{2\pi}{T}$ sampling freq. (radians/sec)

Assume $\Omega_s > 2\Omega_m$ (Nyquist rate)




Reconstruction filter

$$H_r(\Omega) = \begin{cases} T ; & |\Omega| \le \frac{\Omega_s}{2} \\ 0 ; & \text{else} \end{cases}$$

$x_r(t) = x_c(t)$ as long as Nyquist sampling condition is met.

Impulse response for ideal reconstruction $h_r(t)$ = Inverse F.T. of $H_r(\Omega)$

$$= \frac{1}{2\pi} \int_{-\frac{\Omega_s}{2}}^{\Omega_s/2} T\, e^{j\Omega t}\, d\Omega$$

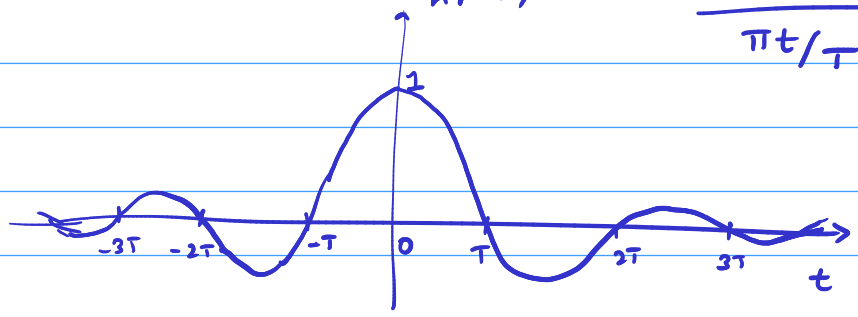
$\Omega_s = \frac{2\pi}{T}$

$\frac{\Omega_s}{2} = \pi/T$

$$= \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} T\, e^{j\Omega t}\, d\Omega$$

$$= \frac{\sin(\pi t/T)}{(\pi t/T)}$$

$$h_r(t) = \frac{\sin \pi t/T}{\pi t/T}$$




$$x_r(t) = x_s(t) * h_r(t)$$

(* : convolution)

$$= \left( \sum_{n=-\infty}^{\infty} x(n)\, \delta(t - nT) \right) * h_r(t)$$

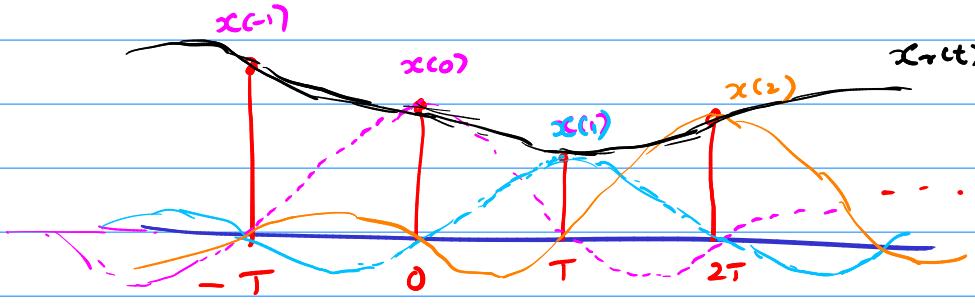
$$= \sum_{n=-\infty}^{\infty} x(n) \left( \delta(t - nT) * h_r(t) \right)$$

$\delta(t-nT) * h_r(t) = h_r(t-nT)$

$$= \sum_{n=-\infty}^{\infty} x(n)\, h_r(t - nT)$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(n) \frac{\sin \frac{\pi}{T}(t - nT)}{\frac{\pi}{T}(t - nT)}$$

Sine Interpolation formula




$x_r(t) = x_c(t)$ if Nyquist Condition met

Ideal reconstruction requires entire sequence $\{x(n)\}$ even to reconstruct $x_c(t)$ at a specific $t$.